Project Proposal: Bayesian Estimation

Stella Rovazzi · Ramez Elprince · Brenden Doogan

University of Guelph

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1 Project Overview

Our group will focus on understanding, briefly explaining, and producing a short report on Bayesian estimation. The framework assumes that parameters of interest are random with probability distributions that describe their uncertainty conditional on the observed dataset [8]. It also treats the dataset as fixed, unlike a frequentist framework where the parameters are fixed, and the data is assumed to be random and a result of repeated sampling from a larger population.

Our computational method of choice for Bayesian estimation is the Gibbs sampling method, which is a part of the Markov Chain Monte Carlo family. MCMC acts as the engine for Bayesian estimation, since it's not always possible to mathematically calculate the posterior distribution. Gibbs sampling is useful when the joint probability distribution $P(X, Y)$ is difficult to sample from directly, but sampling from the conditional distributions, $P(X|Y)$ and $P(Y|X)$ is feasible [1]. We will start with a simple normal model that has known theoretical properties, and we want to see if our computational sampling method is approaching the expected distribution; in a sense, we're looking to see if the history of the random walk traces out the probability distribution of interest. We then plan to explore more advanced models.

The spirit of the project is to quantify how accurately these computational methods help us converge the posterior.

Objectives

Simulation Study

We start off with a straightforward example using a normal model and we see how our simulation results compare to the theoretical results. After establishing the basic framework we extend the application to real data, then we explore how we can extend the methodology for hierarchical models.

Evaluating the Posterior

We will assess the means, variances, and credible intervals of posterior distributions [3]. Discuss how simulation error and sample size influence our results. Then, find the maximum a posteriori [10] [8].

Assessing the Convergence

We will also investigate the convergence of our estimated posteriors using three methods: the Gelman-Rubin statistic $\sqrt{\hat{R}}$ [6] [8], the gibbs stopper method [9], burn-in range [4], and test multiple initializations.

2 Background and Literature

Many of our sources are in the in-text citaitons and mentioned in the bibliography.

- Bayesian Data Analysis, Gelman et al. [3]
  This is the main source for the examples and tutorials we are following. Gelman also introduces the three-step method to Bayesian inference:
  1. Set up the full probability model by getting the joint distribution
  2. Condition on the observed data by getting the posterior, or find the conditional probability of the unobserved quantities of the dataset
  3. Evaluate the fit of the posterior
- Renard Paper Renard et al. [6]
  We can use Bayesian inference in order to assess the suitability of a proposed model for a dynamic system,

$$P(M_j|X) = \frac{P(X|M_j)\pi(M_j)}{\sum_{i=1}^{q} P(X|M_i)\pi(M_i)}$$

  where M_k is the kth proposed model, q is the number of models we propose, and $\pi(M_k)$ is the probability of using model M_k. It is important to note that M_i and M_j are not neccessarily of different distributions, but perhaps different parameters chosen for the same distribution. $P(X|M) = \int P(X, \theta|M)d\theta$ is extremely difficult to solve numerically for high dimensions. However, it acts as a normalizing factor and we can instead sample $P(X|M_j)\pi(M_j)$ using MCMC methods. We can estimate the normalizing factor via the eCDF, as without doing so would cause our samples to be improperly scaled.
  Comparing between models can be done via Bayes factors, where we compare the ith and jth model using:

$$B_{i,j} = \frac{P(X|M_i)}{P(X|M_j)}$$

  where the usage of model M_i is supported by high values of $B_{i,j}$.
  In this application, the Markov chain is comprised of iteratively selecting different parameters (denoted as θ^τ where τ is the τ'th vector of parameters generated). To quantify the convergence of an MCMC model across different starting points, we can use the Gelman-Rubin statistic. The statistic is a measurement of variability between chains, and is further

3 Computational Strategy

The quantities of interest are mentioned in the project overview.

3.1 Required Libraries

- MASS for the bivariate normal generation [7]
- Stan for the hierarchical models (if time permits) [5]

3.2 Simple Example

Goal: Ensure we can converge to a known model using Bayesian techniques.

Methodology: This follows the example C.3 regarding the normals in Bayesain Data Analysis (BDA) [3].

1. Generate bivariate normal data
2. Derive form of conditionals from the joint and implement as R functions
3. Sample from posterior with Gibbs Sampler
4. Assess the if sampled posterior is reasonable
5. Retrieve the other quantites of interest

3.3 Harder Example

Goal: Apply what was learned in the simple example to real data.

Methodology: This follows the bioassay expirement in example 3.7 from BDA [3].

1. Derive the full conditional, then the conditionals using the distributions provided in text
2. Sample from posterior with Gibbs Sampler
3. Assess the if sampled posterior is reasonable
4. Retrieve the other quantites of interest

Dataset This dataset is shown in table 3.1 from the BDA text [3]. The data is binomially distributed.

3.4 Extension: Hierarchical Models

Goal: If time permits, we would use the rat tumor example (5.4) from BDA and tutorial C.3 to try to construct a Bayesian hierarchical model using stan [3], giving us opportunity to learn hyperparameters and Stan.

Methodology:

1. Create stan file, laying out parameters, model, and transformed parameters
2. Run some number of chains iteratively in R, which can be set to be the Gibbs method
3. Assess the if sampled posterior is reasonable
4. Retrieve the other quantites of interest

Dataset: This dataset can be found at Gelman [2] titled "Rat tumors". The data represents the count of rats developing tumours from various expirements (or samples). Y representing the number of rats developing tumors and N representing the total number of rats in the study. The data is binomially distributed.

Division of Responsibility

Report contributions

Person	Responsibility
Ramez	Motivation for Bayesian inference.
Stella	Simulation implementation and computational strategy.
Brenden	Real-world applications and uses of the workflow in hierarchical models.

Presentation topics

These are speaking assignments; preparation may be shared.

Ramez: Motivation and methods
Motivate Bayesian statistics and explain how it differs from frequentist methods.
Explain Gibbs sampling and how it explores possible model parameter values.
Discuss how to assess whether one model is better than another.

Stella: Comparison and simulation
Compare MLE with Bayesian inference, including limitations of MLE and how Bayesian inference can address them.
Present the model through a simple simulation.
Visualize sampled parameter values moving through parameter space.
Consider the example in BDA, Appendix C.3.

Brenden: Diagnostics and applications
Show how parameter values are updated in the sampling workflow.
Demonstrate the Gelman–Rubin statistic using the team's simulated Markov chains and explain what it measures.
Present real-world applications of the workflow.
Introduce hierarchical models and explain how the workflow applies to them.

Timeline

All dates are in 2026.

Deadline	Deliverable
6 Nov.	Submit the proposal.
13 Nov.	**Ramez:** Describe Bayesian inference; compare it with MLE and explain when to use each. **Stella:** Create a simple simulation for the report and presentation. **Brenden:** Find real-world applications; define hierarchical models and identify their use in studies.
20 Nov.	**All:** Complete slides for each presentation section.
22 Nov.	**All:** Assemble and complete the presentation, allowing time to practise.
24 Nov.	Presentations begin.
To be set	**All:** Have the report near completion and ready for editing.
4 Dec.	Submit the final report.

Expected Outcomes

We would expect our model to be able to approximately sample from the wanted distribution through MCMC methods, and be able to estimate the model parameters for further analysis. We would expect to be able to find quantities of interest using our simulation. Depending on how much time we have, we may or may not pursue learning about and implementing a hierarchical model. While we do not necessarily intend to focus much if at all on them in the presentation, they are a valuable tool to be aware of and know how to implement.

References

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[4] Charles Geyer. MCMC/Burn. URL https://users.stat.umn.edu/~geyer/mcmc/burn.html.

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[8] Maria L. Rizzo. *Statistical computing with R*. CRC Press, Taylor & Francis Group, Boca Raton, second edition edition, 2019. ISBN 978-1-4665-5332-3 978-1-4665-5333-0.

[9] Martin A. Tanner. *Tools for Statistical Inference: Methods for the Exploration of Posterior Distributions and Likelihood Functions*. Springer Series in Statistics. Springer New York, New York, NY, 1996. ISBN 978-1-4612-8471-0 978-1-4612-4024-2. doi: 10.1007/978-1-4612-4024-2. URL http://link.springer.com/10.1007/978-1-4612-4024-2.

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